Understanding the Riemann Hypothesis

Sriram Rajagopalan

August 6, 2026

Contents

Preface

This document contains a step-by-step approach to understanding the Riemann hypothesis. I start from the Riemann zeta function and end at the zeros of its functional equation, whose non-trivial zeros, per Riemann's hypothesis, will provide insights into the distribution of prime numbers. I independently studied complex analysis from Complex Variables and Applications by Churchill and Brown. This document is an attempt to use whatever I've learned to understand the Riemann hypothesis - something I've heard of for years, but never had the math to understand it comprehensively. This document is structured as follows: Sections 1 and 2 express the zeta function as a product of the primes. I have also used this to prove the infinitude of primes. Section 3 is a prelude to analytical continuation, which will be used to "complete" the zeta function in Section 5. Section 4 develops the Gamma function, crucial to the "completed zeta function", which is the crown jewel - the function whose zeros are supposedly the key to understanding the distribution of the primes. Some of the proofs apparently are basic analytical number theory, but my knowledge is limited to analysis and calculus. I have primarily used Chapters 6 and 7 of Stein and Shakarchi's Complex Analysis in Sections 5 and 6, along with a mixture of online sources for the derivations in the previous sections. I have handwritten notes for every section and calculation, and I uploaded them to ChatGPT and Claude, which assisted me with the Overleaf code for this document.

Section 7 completes the chain - it establishes the connection between the hypothesis and the Prime Number Theorem, and why the real part of the non-trivial zeros are so important. Unfortunately, the math is beyond my scope(for now), delving deep into analytical number theory. As such, it is treated as an expository section. It reveals the grand truth but does not show the path. Let it be noted that this document was begun on 30 June 2026.

1 The Euler Product Formula

Definition 1.1 (Riemann zeta function)**.** For $s \in \mathbb{C}$ the *Riemann zeta function* is the Dirichlet series

$$\zeta(s) \;=\; \sum_{n=1}^{\infty} \frac{1}{n^s},$$

i.e. the series $\sum_{n\geq 1} a_n n^{-s}$ with $a_n \equiv 1$. The series converges absolutely on the half-plane $\mathrm{Re}(s) > 1$ (Proposition 1.3), and it is on this half-plane that everything below takes place.

1.1 Derivation via the Sieve of Eratosthenes

We work throughout in $\mathrm{Re}(s) > 1$, where $\sum_n n^{-s}$ converges absolutely. Absolute convergence is essential as the following manipulations would otherwise be rendered moot.

Multiplying $\zeta(s)$ by 2^{-s} collects exactly the even-indexed terms, since $2^{-s}\zeta(s) = \sum_{n\geq 1}(2n)^{-s}$. Subtracting therefore sieves out every multiple of 2:

$$\left(1-2^{-s}\right)\zeta(s) \;=\; \sum_{\substack{n\geq 1\\ 2\nmid n}} \frac{1}{n^s} \;=\; 1 + \frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} + \cdots.$$

Multiplying next by $\left(1-3^{-s}\right)$ removes the surviving multiples of 3:

$$\left(1-3^{-s}\right)\left(1-2^{-s}\right)\zeta(s) \;=\; 1 + \frac{1}{5^s} + \frac{1}{7^s} + \frac{1}{11^s} + \cdots.$$

Iterating over the primes $p_1 < p_2 < \cdots < p_k$ in increasing order, each new factor $\left(1-p_i^{-s}\right)$ deletes the terms whose index is divisible by p_i, leaving

$$\prod_{i=1}^{k}\left(1-p_i^{-s}\right)\zeta(s) \;=\; 1 + \sum_{\substack{n>1\\ p\mid n \,\Rightarrow\, p>p_k}} \frac{1}{n^s}. \tag{1}$$

Every index n surviving in the sum on the right has all prime factors exceeding p_k, so in particular $n > p_k$. Hence that sum is bounded by the tail $\sum_{n>p_k} n^{-\sigma}$ (with $\sigma = \mathrm{Re}(s) > 1$), which tends to 0 as $k \to \infty$ because it is the tail of a convergent series. Letting $k \to \infty$ in (1) therefore gives $\prod_p\left(1-p^{-s}\right)\zeta(s) = 1$.

Theorem 1.2 (Euler product)**.** *For* $\mathrm{Re}(s) > 1$*,*

$$\zeta(s) \;=\; \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}.$$

Remark. The step "repeating this infinitely we obtain 1" is exactly the limit $k \to \infty$ above, and what makes it rigorous is that the leftover terms in (1) all have index $> p_k$, so they form the tail of a convergent series[1].

[1] This also shows ζ is zero-free on $\mathrm{Re}(s) > 1$. $\prod_{i=1}^{k}\left(1-p_i^{-s}\right)\zeta(s) = 1 + \sum_{\substack{n>1\\ p\mid n \,\Rightarrow\, p>p_k}} n^{-s}$, and the right-hand side tends to 1 as $k \to \infty$. Were $\zeta(s) = 0$, the left-hand side would vanish identically in k, forcing $0 = 1$.

1.2 Convergence of the Zeta Function

Proposition 1.3 (Region of convergence)**.** *The series* $\sum_{n\geq 1} n^{-s}$ *converges absolutely if and only if* $\mathrm{Re}(s) > 1$.

Proof. Write $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. For each $n \geq 1$,

$$\left|n^{-s}\right| = \left|n^{-\sigma}\right| \left|n^{-it}\right| = n^{-\sigma} \left|\cos(t \ln n) - i \sin(t \ln n)\right| = n^{-\sigma},$$

since $n^{-it} = e^{-it \ln n}$ has modulus 1. Consequently $\sum_n \left|n^{-s}\right| = \sum_n n^{-\sigma}$, which is a real p-series and converges when $\sigma = \mathrm{Re}(s) > 1$. □

Remark. At $\sigma = 1$ exactly, the series is the harmonic series $\sum_n 1/n$, which diverges; so the boundary $\mathrm{Re}(s) = 1$ is excluded. This divergence is exactly what produces the pole of ζ at $s = 1$, used in the next subsection.

2 Infinitude of the Primes

The Euler product turns a statement about primes into a statement about ζ near $s = 1$. Taking logarithms of the product (valid for real $s > 1$, where every factor is positive) and expanding $-\log(1 - x) = \sum_{k\geq 1} x^k/k$,

$$\log \zeta(s) = \sum_p -\log\bigl(1 - p^{-s}\bigr) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k\, p^{ks}} = \underbrace{\sum_p \frac{1}{p^s}}_{\text{the } k=1 \text{ terms}} + \underbrace{\sum_p \sum_{k=2}^{\infty} \frac{1}{k\, p^{ks}}}_{=: R(s)}.$$

The remainder $R(s)$ is *bounded* as $s \to 1^+$: estimating crudely,

$$R(s) \;\leq\; \sum_p \sum_{k=2}^{\infty} p^{-ks} \;=\; \sum_p \frac{p^{-2s}}{1 - p^{-s}} \;\leq\; 2 \sum_p p^{-2s} \;\leq\; 2 \sum_{n=2}^{\infty} n^{-2} \;<\; \infty,$$

uniformly for $s \geq 1$ (using $1 - p^{-s} \geq \frac{1}{2}$). Write $R(s) = O(1)$ for this bound.

It remains to show $\log \zeta(s) \to \infty$ as $s \to 1^+$. Comparing the series to an integral (the integrand x^{-s} is positive and decreasing, so $\sum_{n\geq 1} f(n) \geq \int_1^{\infty} f$),

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \;\geq\; \int_1^{\infty} \frac{dx}{x^s} \;=\; \frac{1}{s-1} \;\xrightarrow[s \to 1^+]{}\; \infty,$$

hence $\log \zeta(s) \to \infty$. Combining this with the boundedness of $R(s)$,

$$\sum_p \frac{1}{p^s} \;=\; \log \zeta(s) - R(s) \;\geq\; \log \zeta(s) - O(1) \;\xrightarrow[s \to 1^+]{}\; \infty. \tag{2}$$

Theorem 2.1. *There are infinitely many primes.*

Proof. If the primes were finite in number, then $\sum_p p^{-s}$ would be a finite sum of terms each bounded by 1, hence bounded as $s \to 1^+$. This contradicts (2). □

Remark. The divergence in (2) requires a *lower* bound on $\sum_p p^{-s}$, and the inequality must run $\sum_p p^{-s} \geq \log \zeta(s) - O(1)$. The fact doing the work is that the $k \geq 2$ remainder $R(s)$ is *bounded*; its mere positivity would only give the (useless) upper bound $\sum_p p^{-s} \leq \log \zeta(s)$.

3 Analytic Continuation via the Eta Function

The zeta series converges only on $\mathrm{Re}(s) > 1$, so to reach smaller $\mathrm{Re}(s)$ we need a different representation. The device is an auxiliary series whose alternating signs let it converge further to the left; matching it against ζ on the overlap then carries ζ along with it.

Definition 3.1 (Dirichlet eta function)**.** For $s \in \mathbb{C}$ the *Dirichlet eta function* is

$$\eta(s) \;=\; \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s} \;=\; 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \cdots .$$

Proposition 3.2 (Domain of the eta series)**.** *The series defining $\eta(s)$ converges, and represents a holomorphic function, on the half-plane* $\mathrm{Re}(s) > 0$.

Proof. Apply Dirichlet's test[2](bounded-variation form) to $\sum a_n b_n$ with $a_n = (-1)^{n+1}$ and $b_n = n^{-s}$, where $\sigma := \mathrm{Re}(s) > 0$. The partial sums $A_k = \sum_{n=1}^{k} a_n \in \{0, 1\}$ are bounded. For the second sequence, $|b_n| = n^{-\sigma} \to 0$ by the modulus identity of proposition 1.3, and b_n has bounded variation:

$$\sum_{n\geq 1} |b_{n+1} - b_n| = \sum_{n\geq 1} \left| s \int_n^{n+1} x^{-s-1}\, dx \right| \leq |s| \int_1^{\infty} x^{-\sigma-1}\, dx = \frac{|s|}{\sigma} < \infty.$$

Dirichlet's test then yields convergence. The bound $|s|/\sigma$ is uniform on compact subsets of $\{\mathrm{Re}(s) > 0\}$, so the convergence is locally uniform and η is holomorphic there by the Weierstrass theorem [3]. □

Remark. The whole gain lives in the alternating signs: they make the partial sums A_k bounded, which is what lets η reach $\mathrm{Re}(s) > 0$, a full unit to the left of ζ's own series. The boundary is sharp. On $\mathrm{Re}(s) = 0$ the terms n^{-s} have modulus 1 and do not tend to 0, so the series *diverges* there - in particular at $s = 0$. This is exactly why the representation reaches $\mathrm{Re}(s) > 0$ and no further; pushing past the imaginary axis will later require the functional equation, not this series (the *function* η does continue further; it is only this representation of it that stops at the boundary).

Lemma 3.3 (Relation between η and ζ)**.** *For* $\mathrm{Re}(s) > 1$,

$$\eta(s) \;=\; \left(1 - 2^{1-s}\right) \zeta(s).$$

[2]The conditions of the test are $\sum a_n$ being bounded and $b_n \to 0$ with bounded variation, $\sum_n |b_{n+1} - b_n| < \infty$. The variation is used as s is a complex number, which makes n^{-s} not monotonic; therefore we use bounded variation.

[3]The theorem states that the limit or sum of holomorphic functions converges to a holomorphic function on a domain if the convergence is uniform on compact subsets.

Proof. On $\mathrm{Re}(s) > 1$ both series converge absolutely, which licenses the termwise subtraction

$$\zeta(s) - \eta(s) = \sum_{n=1}^{\infty} \frac{1 - (-1)^{n+1}}{n^s} = \sum_{n=1}^{\infty} \frac{1 + (-1)^n}{n^s}.$$

The numerator vanishes for odd n and equals 2 for even n, so only the even indices $n = 2m$ survive:

$$\zeta(s) - \eta(s) = 2 \sum_{m=1}^{\infty} \frac{1}{(2m)^s} = 2^{1-s} \sum_{m=1}^{\infty} \frac{1}{m^s} = 2^{1-s}\,\zeta(s).$$

Hence $\eta(s) = \zeta(s) - 2^{1-s}\zeta(s) = (1 - 2^{1-s})\,\zeta(s)$. $\square$

Theorem 3.4 (Continuation to $\mathrm{Re}(s) > 0$). *The function*

$$\zeta(s) \;=\; \frac{\eta(s)}{1 - 2^{1-s}}$$

is the analytic continuation of ζ to the half-plane $\mathrm{Re}(s) > 0$, holomorphic except for a simple pole at $s = 1$.

Proof. By proposition 3.2 the numerator is holomorphic on $\mathrm{Re}(s) > 0$. The denominator is entire, and $1 - 2^{1-s} = 0$ exactly when $2^{1-s} = 1$, i.e. $(1 - s)\ln 2 \in 2\pi i$, that is

$$s = 1 + \frac{2\pi i k}{\ln 2}, \qquad k \in \mathbb{Z},$$

a discrete set lying on the line $\mathrm{Re}(s) = 1$. Off this set the quotient is holomorphic on $\mathrm{Re}(s) > 0$, and by lemma 3.3 it coincides with ζ on $\mathrm{Re}(s) > 1$. That half-plane has limit points, so by the identity theorem the quotient is the unique analytic continuation of ζ — existence supplied by the formula, uniqueness by the identity theorem.

At $s = 1$ the denominator has a simple zero.[4] $\square$

4 The Gamma Function

The gamma function is the analytic scaffolding for the functional equation of the next section. Here we define it, establish holomorphy on $\mathrm{Re}(s) > 0$, prove the recursion it satisfies, and use that recursion to continue it to a meromorphic function on all of $\mathbb{C}$. The poles it acquires, and their residues, are exactly what later manufacture the trivial zeros of ζ.

Definition 4.1 (Gamma function). For $\mathrm{Re}(s) > 0$,

$$\Gamma(s) \;=\; \int_0^{\infty} t^{s-1} e^{-t}\, dt.$$

Proposition 4.2 (Convergence). *The integral converges for $\mathrm{Re}(s) > 0$, uniformly on every closed strip $0 < b \le \mathrm{Re}(s) \le a$.*

[4]The formula also has apparent singularities at $s = 1 + 2\pi i k/\ln 2$ for $k \neq 0$. These turn out to be *removable* as ζ is holomorphic there

Proof. Write $\sigma = \mathrm{Re}(s)$ and split the integral at $t = 1$. The integrand has modulus $|t^{s-1}e^{-t}| = t^{\sigma-1}e^{-t}$.

Near ∞. For $\sigma \le a$ and $t \ge 1$, $t^{\sigma-1} \le t^{a-1}$, so $t^{\sigma-1}e^{-t} \le t^{a-1}e^{-t}$, and $\int_1^\infty t^{a-1}e^{-t}\,dt < \infty$ because e^{-t} decays faster than any power grows.

Near 0. For $\sigma \ge b$ and $0 < t \le 1$, $t^{\sigma-1} \le t^{b-1}$, so $t^{\sigma-1}e^{-t} \le t^{b-1}$, and $\int_0^1 t^{b-1}\,dt = \frac{1}{b} < \infty$ since $b > 0$.

Hence, in the strip $0 < b <\mathrm{Re}(s)< a$, the integrand is bounded by an integrable function independent of s. Therefore, the integral converges uniformly on the strip. □

Proposition 4.3. Γ *is holomorphic on* $\{\mathrm{Re}(s) > 0\}$*, with* $\Gamma'(s) = \displaystyle\int_0^\infty t^{s-1}e^{-t}\log t\,dt$.

Proof. We will show that the partial derivative with respect to s is uniformly convergent, which allows differentiating under the integral sign. This will prove that Γ is complex differentiable and therefore holomorphic.
Fix a strip $0 < b \le \mathrm{Re}(s) \le a$. $\partial_s\big(t^{s-1}e^{-t}\big) = t^{s-1}e^{-t}\log t$, of modulus $t^{\sigma-1}e^{-t}|\log t|$. Since $|\log t|$ is dominated by any small power of t as $t \to 0^+$ and by $e^{t/2}$ as $t \to \infty$, the same bounds as in proposition 4.2 apply, so the differentiated integral also converges uniformly on the strip. Now, differentiating under the integral sign gives the stated formula for $\Gamma'(s)$; in particular Γ is complex-differentiable, hence holomorphic, on $\mathrm{Re}(s) > 0$. □

Proposition 4.4 (Functional equation)**.** $\Gamma(1) = 1$*, and for* $\mathrm{Re}(s) > 0$*,*

$$\Gamma(s+1) = s\,\Gamma(s).$$

Consequently $\Gamma(n) = (n-1)!$ *for every* $n \in \mathbb{Z}^+$.

Proof. Directly, $\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$. For the recursion, integrate by parts with $u = t^s$, $dv = e^{-t}\,dt$:

$$\Gamma(s+1) = \int_0^\infty t^s e^{-t}\,dt = \Big[-t^s e^{-t}\Big]_0^\infty + s\int_0^\infty t^{s-1}e^{-t}\,dt.$$

The first term reduces to 0. Hence $\Gamma(s+1) = s\,\Gamma(s)$. Combined with $\Gamma(1) = 1$, induction gives $\Gamma(n) = (n-1)!$. □

4.1 Meromorphic continuation

Theorem 4.5 (Meromorphic continuation)**.** Γ *extends to a meromorphic*[5] *function on* $\mathbb{C}$*, holomorphic except for simple poles at* $s = 0, -1, -2, \ldots$*. On* $\{\mathrm{Re}(s) > -m\} \setminus \{0, -1, \ldots, -(m-1)\}$ *it is given by*

$$\Gamma(s) \;=\; \frac{\Gamma(s+m)}{s(s+1)(s+2)\cdots(s+m-1)}.$$

Proof. Take $m = 1$ first. Define $\Gamma_1(s) = \Gamma(s+1)/s$ on $\{\mathrm{Re}(s) > -1\}$. The numerator is holomorphic there (as $\mathrm{Re}(s+1) > 0$), and the denominator vanishes only at $s = 0$, so Γ_1 is holomorphic on $\{\mathrm{Re}(s) > -1\} \setminus \{0\}$ with a simple pole at $s = 0$. By proposition 4.4, $\Gamma_1 = \Gamma$

[5]Meromorphic functions are defined as functions that are holomorphic at all points in its domain except at its poles

on $\{\mathrm{Re}(s) > 0\}$; that half-plane has limit points, so by the Identity theorem Γ_1 is the unique analytic continuation of Γ to $\{\mathrm{Re}(s) > -1\}$.

Iterating the recursion m times gives $\Gamma(s+m) = s(s+1)\cdots(s+m-1)\,\Gamma(s)$ on $\{\mathrm{Re}(s) > 0\}$, and the same argument shows

$$\Gamma_m(s) = \frac{\Gamma(s+m)}{s(s+1)\cdots(s+m-1)}$$

continues Γ to $\{\mathrm{Re}(s) > -m\}$: the numerator is holomorphic and nonzero there, while the denominator contributes simple poles exactly at $s = 0, -1, \ldots, -(m-1)$. As m is arbitrary, Γ is meromorphic on $\mathbb{C}$ with simple poles at the nonpositive integers. □

Proposition 4.6 (Residues at the poles)**.** *For each $n = 0, 1, 2, \ldots,$*

$$\operatorname*{Res}_{s=-n} \Gamma(s) \;=\; \frac{(-1)^n}{n!}.$$

Proof. Choose any $m > n$, so that $s = -n$ lies in $\{\mathrm{Re}(s) > -m\}$. Near $s = -n$ isolate the vanishing factor and bring it to the residue form[6]

$$\phi(s) = \frac{\Gamma(s+m)}{\displaystyle\prod_{\substack{0\le j\le m-1\\ j\ne n}} (s+j)}, \qquad \Gamma(s) = \frac{\phi(s)}{s+n},$$

the pole is simple, so its residue is $\phi(-n) = \dfrac{\Gamma(m-n)}{\prod_{j\ne n}(j-n)}$. Split the product at $j = n$:

$$\prod_{j=0}^{n-1}(j-n) = (-n)(-(n-1))\cdots(-1) = (-1)^n n!, \qquad \prod_{j=n+1}^{m-1}(j-n) = 1\cdot 2\cdots(m-1-n) = (m-1-n)!.$$

Since $\Gamma(m-n) = (m-n-1)!$,

$$\operatorname*{Res}_{s=-n} \Gamma(s) = \phi(-n) = \frac{(m-n-1)!}{(-1)^n\, n!\,(m-1-n)!} = \frac{(-1)^n}{n!},$$

the two equal factorials cancel. The value is independent of the m, as it must be. □

Remark. These simple poles are the whole point of the section for what follows. In the completed zeta function $\xi(s)$ the factor $\Gamma(s/2)$ carries exactly this pole structure, and its poles are what force the trivial zeros of ζ at the negative even integers.

5 Completing the Zeta Function

The functional equation is the heart of the theory: it reflects the plane about the line $\mathrm{Re}(s) = \frac{1}{2}$ and, in doing so, fixes the location of the trivial zeros and frames the question of the nontrivial ones. We first close the account on Γ with the reflection formula, which shows Γ is zero-free, and then build the completed zeta function $\xi(s)$ and prove its symmetry.

[6] $\Gamma(s) = \frac{\phi(s)}{(s-s_0)^n}$

Theorem 5.1 (S&S Theorem 1.4)**.** *For $s \in \mathbb{C}$,*

$$\Gamma(s)\,\Gamma(1-s) = \frac{\pi}{\sin \pi s}.$$

Proof. Take first $0 < \mathrm{Re}(s) < 1$. Writing the two factors as integrals and substituting $u = vt$ in the second,

$$\Gamma(s)\Gamma(1-s) = \int_0^\infty \int_0^\infty e^{-t} t^{s-1}\, e^{-vt}(vt)^{-s}\, t\, dv\, dt = \int_0^\infty \int_0^\infty e^{-t(1+v)} v^{-s}\, dt\, dv,$$

the powers of t cancelling to t^0. The double integral converges absolutely on this strip, so Fubini's theorem applies. Performing the inner t-integral, $\int_0^\infty e^{-t(1+v)}\, dt = (1+v)^{-1}$, leaves

$$\Gamma(s)\Gamma(1-s) = \int_0^\infty \frac{v^{-s}}{1+v}\, dv.$$

Using $\int_0^\infty \frac{v^{a-1}}{1+v}\, dv = \frac{\pi}{\sin \pi a}$ for $0 < a < 1$ (S&S), taken with $a = 1-s$ and $v = e^x$, this is $\pi/\sin\pi(1-s) = \pi/\sin\pi s$. Both sides are meromorphic and agree on the strip, so by the Identity theorem the formula holds for all $s \in \mathbb{C}$. □

Remark. This closes the zero-freeness of Γ. If $\Gamma(s_0) = 0$ the left side would vanish, but the right side $\pi/\sin\pi s_0$ never does; the only way to reconcile a zero would be a compensating pole of $\Gamma(1-s_0)$, which occurs only at $s_0 \in \mathbb{Z}^+$ — and there $\Gamma(s_0)$ is a nonzero factorial. Hence Γ has no zeros anywhere.

5.1 The Theta Function and the Completed Zeta

Definition 5.2 (Theta function)**.** For $u > 0$,

$$\vartheta(u) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 u}, \qquad \text{so} \qquad \psi(u) := \sum_{n=1}^{\infty} e^{-\pi n^2 u} = \frac{\vartheta(u)-1}{2}.$$

Theorem 5.3 (S&S Theorem 2.2)**.** *For* $\mathrm{Re}(s) > 1$*,*

$$\pi^{-s/2}\,\Gamma\big(\tfrac{s}{2}\big)\,\zeta(s) = \frac{1}{2}\int_0^\infty u^{s/2-1}\big[\vartheta(u)-1\big]\, du.$$

Proof. Substituting $u = t/(\pi n^2)$ gives, for each $n \ge 1$,

$$\int_0^\infty e^{-\pi n^2 u}\, u^{s/2-1}\, du = (\pi n^2)^{-s/2} \int_0^\infty e^{-t} t^{s/2-1}\, dt = \pi^{-s/2}\,\Gamma\big(\tfrac{s}{2}\big)\, n^{-s}.$$

Summing over $n \ge 1$ and using $\sum_{n\ge1} n^{-s} = \zeta(s)$,

$$\pi^{-s/2}\Gamma\big(\tfrac{s}{2}\big)\,\zeta(s) = \sum_{n=1}^{\infty}\int_0^\infty e^{-\pi n^2 u} u^{s/2-1}\, du = \int_0^\infty u^{s/2-1} \sum_{n=1}^{\infty} e^{-\pi n^2 u}\, du = \int_0^\infty u^{s/2-1}\psi(u)\, du.$$

The interchange of sum and integral is justified by the decay of ψ (the bounds are those mentioned in S&S). Finally $\psi(u) = (\vartheta(u)-1)/2$ gives the stated form. □

Definition 5.4 (Completed zeta function)**.**

$$\xi(s) = \pi^{-s/2}\,\Gamma\big(\tfrac{s}{2}\big)\,\zeta(s).$$

Theorem 5.5. *For $u > 0$, the theta function satisfies*

$$\vartheta(u) = u^{-1/2}\,\vartheta(1/u).$$

This is the Poisson summation formula applied to the Gaussian; we take it as a known result and do not reproduce the proof.

The form actually used below is the corresponding identity for $\psi(u) = \frac{1}{2}(\vartheta(u) - 1)$. Since $\vartheta(u) = u^{-1/2}\vartheta(1/u)$,

$$\psi(u) = \frac{\vartheta(u) - 1}{2} = \frac{u^{-1/2}\vartheta(1/u) - 1}{2} = u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}.$$

Theorem 5.6 (S&S Theorem 2.3 — functional equation)**.** *ξ is holomorphic for* $\mathrm{Re}(s) > 1$ *and continues to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at $s = 0$ and $s = 1$. Moreover*

$$\xi(s) = \xi(1 - s) \qquad \textit{for all } s \in \mathbb{C}.$$

Proof. By theorem 5.3, $\xi(s) = \int_0^\infty u^{s/2-1}\psi(u)\,du$ for $\mathrm{Re}(s) > 1$. Split at $u = 1$. On $(0, 1)$ substitute the theta relation theorem 5.5, $\psi(u) = u^{-1/2}\psi(1/u) + \frac{1}{2}u^{-1/2} - \frac{1}{2}$, and then send $u \mapsto 1/u$ in the resulting $\psi(1/u)$ term.

$$\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \int_0^\infty u^{\frac{s}{2}-1}\psi(u)\,du.$$

Split the integral at $u = 1$:

$$\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \int_0^1 u^{\frac{s}{2}-1}\psi(u)\,du + \int_1^\infty u^{\frac{s}{2}-1}\psi(u)\,du.$$

Using the functional equation for ψ,

$$\begin{aligned}\int_0^1 u^{\frac{s}{2}-1}\psi(u)\,du &= \int_0^1 u^{\frac{s}{2}-1}\left(u^{-1/2}\psi\left(\frac{1}{u}\right) + \frac{1}{2u^{1/2}} - \frac{1}{2}\right)du \\ &= \int_0^1 u^{\frac{s}{2}-\frac{3}{2}}\psi\left(\frac{1}{u}\right)du + \frac{1}{2}\int_0^1 u^{\frac{s}{2}-\frac{3}{2}}\,du - \frac{1}{2}\int_0^1 u^{\frac{s}{2}-1}\,du.\end{aligned}$$

The elementary integrals are

$$\frac{1}{2}\int_0^1 u^{\frac{s}{2}-\frac{3}{2}}\,du = \frac{1}{s-1}, \qquad \frac{1}{2}\int_0^1 u^{\frac{s}{2}-1}\,du = \frac{1}{s}.$$

For the remaining integral, let $v = 1/u$. Then

$$du = -\frac{1}{v^2}\,dv,$$

and hence

$$\int_0^1 u^{\frac{s}{2}-\frac{3}{2}}\psi\left(\frac{1}{u}\right)\,du = \int_\infty^1 \left(\frac{1}{v}\right)^{\frac{s}{2}-\frac{3}{2}}\psi(v)\left(-\frac{1}{v^2}\right)\,dv$$
$$= \int_1^\infty v^{-\frac{s}{2}-\frac{1}{2}}\psi(v)\,dv.$$

Therefore,

$$\boxed{\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty \left(u^{\frac{s}{2}-1} + u^{-\frac{s}{2}-\frac{1}{2}}\right)\psi(u)\,du.}$$

The two elementary pieces integrate to

$$\frac{1}{2}\int_0^1 u^{s/2-3/2}\,du = \frac{1}{s-1}, \qquad -\frac{1}{2}\int_0^1 u^{s/2-1}\,du = -\frac{1}{s},$$

and the substituted term[7] becomes $\int_1^\infty u^{(1-s)/2-1}\psi(u)\,du$. Hence

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty \left(u^{s/2-1} + u^{(1-s)/2-1}\right)\psi(u)\,du. \tag{3}$$

The integral in (3) converges for *every* $s \in \mathbb{C}$ and defines an entire function; the elementary part $\frac{1}{s-1} - \frac{1}{s} = \frac{1}{s(s-1)}$ supplies the simple poles at $s = 0$ and $s = 1$. This is the meromorphic continuation of ξ. Finally, $\xi(s) = \xi(1-s)$ should be obvious to spot. □

5.2 Meromorphic Continuation of ζ

Proposition 5.7. *$\xi(s) \neq 0$ whenever $\mathrm{Re}(s) < 0$.*

Proof. Suppose $\xi(s_0) = 0$ with $\mathrm{Re}(s_0) < 0$. Recall $\xi(1-s_0) = 0$, with $\mathrm{Re}(1-s_0) > 1$. On that half-plane $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, in which $\pi^{-s/2}$ is a nonvanishing exponential, Γ is zero-free (remark following Theorem 5.1), and ζ is nonvanishing by the Euler product on $\mathrm{Re}(s) > 1$. Hence $\xi(1-s_0) \neq 0$, a contradiction. □

Theorem 5.8 (S&S Theorem 2.4)**.** *ζ continues to a meromorphic function on $\mathbb{C}$ whose only singularity is a simple pole at $s = 1$. Its only zeros in $\mathrm{Re}(s) \leq 0$ are the* trivial zeros *- negative even integers $s = -2, -4, -6, \ldots$*

Proof. By the definition of

$$\zeta(s) = \pi^{s/2}\frac{\xi(s)}{\Gamma(s/2)}.$$

By theorem 5.6, ξ is meromorphic on $\mathbb{C}$ with simple poles only at $s = 0, 1$; and $1/\Gamma(s/2)$ is entire (theorem 5.1 shows Γ is zero-free, so $1/\Gamma$ has no poles), with simple zeros exactly where $\Gamma(s/2)$ has poles, namely $s/2 = 0, -1, -2, \ldots$, i.e. $s = 0, -2, -4, \ldots$.

At $s = 1$, ξ has a simple pole and $\Gamma(\frac{1}{2}) = \sqrt{\pi} \neq 0$, so ζ has a simple pole, and ξ does not have any zeros when $\mathrm{Re}(s){<}0$. *At* $s = 0$, the simple pole of ξ is cancelled by the simple zero of $1/\Gamma(s/2)$, so ζ is regular at 0 (and $s = 0$ is not a zero).[8] □

[7] Note that when the change of variables is completed, v is replaced by u since integration variables are essentially dummy variables.

[8] Letting s tend to 0, $\Gamma(s/2)$ can be approximated as $\frac{2}{s}$, which will be cancelled by the $\frac{1}{s}$ in the numerator.

6 The Riemann Hypothesis

The functional equation confines the remaining zeroes in the *critical strip* $0 < \mathrm{Re}(s) \leq 1$, and they occur in the symmetric pattern forced by $\xi(s) = \xi(1-s)$. Riemann conjectured that they all lie on the central line.

> **Riemann Hypothesis**. Every nontrivial zero of $\zeta(s)$(every zero in the critical strip) lies on the line $\mathrm{Re}(s) = \frac{1}{2}$.

The conjecture has not been proved till now. What can be proved is where the zeros *can't* be: the boundary line $\mathrm{Re}(s) = 1$ (and, by the functional equation, $\mathrm{Re}(s) = 0$) carries no zeros at all. That is the content of the rest of this section. $s = 0$ is not a nontrivial zero, since ζ is regular and nonzero there (theorem 5.8).

Lemma 6.1 (S&S Lemma 1.3 - this section refers to Chapter 7)**.** *For* $\mathrm{Re}(s) > 1$,

$$\log\zeta(s) = \sum_{p,\,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n\, n^{-s}, \qquad c_n \geq 0,$$

where $c_n = 1/m$ *if* $n = p^m$ *is a prime power and* $c_n = 0$ *otherwise.*

Proof. Taking the logarithm of the Euler product and expanding $-\log(1-x) = \sum_{m\geq 1} x^m/m$ (valid as $|p^{-s}| < 1$ for $\mathrm{Re}(s) > 1$),

$$\log\zeta(s) = \sum_p -\log\bigl(1 - p^{-s}\bigr) = \sum_p \sum_{m=1}^{\infty} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s},$$

the double sum converging absolutely on $\mathrm{Re}(s) > 1$, so that the regrouping by $n = p^m$ is legitimate. $c_n \geq 0$. $\square$

Lemma 6.2 (S&S Lemma 1.4)**.** *For every* $\theta \in \mathbb{R}$ $\quad 3 + 4\cos\theta + \cos 2\theta \geq 0$.

Proof. Using $\cos 2\theta = 2\cos^2\theta - 1$,

$$3 + 4\cos\theta + \cos 2\theta = 2 + 4\cos\theta + 2\cos^2\theta = 2\bigl(1 + \cos\theta\bigr)^2 \geq 0.$$

$\square$

Corollary 6.3 (S&S Corollary 1.5)**.** *For* $\sigma > 1$ *and* $t \in \mathbb{R}$,

$$\log\bigl|\zeta(\sigma)^3\,\zeta(\sigma+it)^4\,\zeta(\sigma+2it)\bigr| \geq 0, \qquad \textit{equivalently} \qquad \bigl|\zeta(\sigma)^3\zeta(\sigma+it)^4\zeta(\sigma+2it)\bigr| \geq 1.$$

Proof. Since $\log|\zeta(s)| = \mathrm{Re}\log\zeta(s)$, lemma 6.1 gives $\log|\zeta(\sigma+i\tau)| = \sum_n c_n n^{-\sigma}\cos(\tau\log n)$. Hence

$$\log\bigl|\zeta(\sigma)^3\zeta(\sigma+it)^4\zeta(\sigma+2it)\bigr| = \sum_{n=1}^{\infty} c_n\, n^{-\sigma}\bigl[\,3 + 4\cos(t\log n) + \cos(2t\log n)\,\bigr].$$

Each bracket is ≥ 0 by lemma 6.2 (with $\theta = t\log n$) and each $c_n n^{-\sigma} \geq 0$, so the sum is ≥ 0. $\square$

Theorem 6.4 (S&S Theorem 1.2)**.** *$\zeta(s)$ has no zeros on the line* $\mathrm{Re}(s) = 1$*.*

Proof. The point $s = 1$ is a pole, not a zero, so suppose for contradiction that $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. We track the orders of vanishing of the three factors in corollary 6.3 as $\sigma \to 1^+$.

$\zeta(\sigma)^3$. Near $s = 1$, $\zeta(s) = \frac{1}{s-1} + O(1)$, so $|\zeta(\sigma)| \sim (\sigma - 1)^{-1}$ and $|\zeta(\sigma)^3| \sim (\sigma - 1)^{-3}$.

$\zeta(\sigma + it_0)^4$. As ζ is holomorphic at $1 + it_0$ (which is not the pole) and vanishes there, $\zeta(\sigma + it_0) = O(\sigma - 1)$, so $|\zeta(\sigma + it_0)^4| = O\big((\sigma - 1)^4\big)$.

$\zeta(\sigma + 2it_0)$. This is holomorphic at $1 + 2it_0$, hence bounded as $\sigma \to 1^+$: $\zeta(\sigma + 2it_0) = O(1)$.

Multiplying, the order in $(\sigma - 1)$ is at least $-3 + 4 + 0 = 1$, so

$$\big|\zeta(\sigma)^3\zeta(\sigma + it_0)^4\zeta(\sigma + 2it_0)\big| = O(\sigma - 1) \xrightarrow[\sigma \to 1^+]{} 0.$$

This contradicts corollary 6.3, which forces the product to stay ≥ 1. Hence no such zero exists, and ζ has no zeros on $\mathrm{Re}(s) = 1$. □

Remark. The whole force is the arithmetic of the exponents: the pole contributes order -3, a zero at $1+it_0$ contributes order $+4$ (a simple zero already suffices, and a higher-order one only helps), and the third factor contributes 0. Their sum is strictly positive, which is incompatible with a product bounded below by 1. By the functional equation, this also clears the line $\mathrm{Re}(s) = 0$, so the nontrivial zeros lie strictly inside the open strip $0 < \mathrm{Re}(s) < 1$.

7 Why is $Re(s) = \frac{1}{2}$ Important?

The Riemann Hypothesis, at first glance to the uninitiated, might seem loosely connected to prime numbers. Seemingly, the only connection is that the zeta function can be expressed as a product of the primes. But how exactly do the nontrivial zeros of its functional equation tell us the spread of prime numbers? To answer that, one needs to understand the Prime Number Theorem(PNT), which states that the number of primes less than or equal to x, denoted $\pi(x)$, can be approximated:

$\pi(x) \approx \frac{x}{lnx}$.

But how exactly does this relate to the non-trivial zeros having a real part of exactly one-half? In essence, the PNT governs the growth of prime numbers. However, it is asymptotic(error terms exist). So the question arises: how exactly are the fluctuations smoothed out?

To answer that, we begin with the logarithmic derivative of the zeta function. Differentiating the Euler product gives

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s},$$

where $\Lambda(n)$ is the von Mangoldt[9] function, defined by

[9]This piecewise function weighs primes.

$$\Lambda(n) = \begin{cases} \log p, & \text{if } n = p^m \text{ for some prime } p \text{ and } m \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Unlike the zeta function itself, this Dirichlet series is supported only on prime powers, making the arithmetic information much more explicit. We now have a direct connection between the analytic properties of $\zeta(s)$ and the distribution of the prime numbers.

Using Perron's formula and the residue theorem, one can derive the *explicit formula*, which expresses the Chebyshev function

$$\psi(x) = \sum_{n \leq x} \Lambda(n)$$

in terms of the nontrivial zeros of the zeta function:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} + \text{lower-order terms},$$

where the sum extends over all nontrivial zeros ρ of $\zeta(s)$.

This identity is the fundamental bridge between the zeros of the zeta function and the distribution of prime numbers. The first term, x, represents the main growth predicted by the Prime Number Theorem, while each nontrivial zero contributes an oscillatory correction. Defining a zero as

$$\rho = \beta + i\gamma,$$

we have

$$x^{\rho} = x^{\beta} e^{i\gamma \log x}.$$

The imaginary part γ determines the frequency of the oscillation, whereas the real part β determines its magnitude. Zeros with larger real parts produce larger fluctuations in the distribution of the primes.

The Prime Number Theorem follows from the fact that the zeta function has no zeros on the line $Re(s) = 1$. This ensures that every oscillatory contribution grows more slowly than the main term x, yielding

$$\pi(x) \sim \frac{x}{\log x}.$$

The Riemann Hypothesis makes a much stronger assertion that every nontrivial zero satisfies

$$Re(\rho) = \frac{1}{2}.$$

If true, every oscillatory contribution has magnitude $x^{1/2}$, the smallest possible magnitude compatible with the functional equation. Consequently, the error in the Prime Number Theorem is essentially as small as it can be. The connection between RH and PNT can be precisely stated as follows: **while the Prime Number Theorem describes the average growth of prime**

numbers, the Riemann Hypothesis predicts the optimal bound on their fluctuations around this average.

8 References

1. J. W. Brown and R. V. Churchill, *Complex Variables and Applications*, 8th ed., McGraw-Hill, New York, 2009.

2. E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton Lectures in Analysis, vol. II, Princeton University Press, Princeton, NJ, 2003.

3. N. Srivastava, *Analytic Continuation and* Γ, lecture notes, 23 November 2015.

4. D. Meyer, *Euler's Product Formula and the Riemann Zeta Function*, unpublished notes, last updated 19 June 2026.

5. B. Riemann, *On the Number of Prime Numbers less than a Given Quantity* (Über die Anzahl der Primzahlen unter einer gegebenen Größe), Monatsberichte der Berliner Akademie, November 1859; translated by D. R. Wilkins, preliminary version, December 1998.